The One-Dimensional Transverse Ising Model

Alexandra-Daria Dumitriu-Iovanescu

10140120

School of Physics and Astronomy

University of Manchester

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Abstract

This essay presents a few of the key techniques used for quantum Ising models, while focusing primarily on finding the exact solution of the 1D spin-1/2 Ising model in a transverse field. Fermionic creation and annihilation operators are introduced in a detailed manner and then used throughout the essay, for both single and multipartite systems. After introducing and analysing the properties of spin operators, a mapping from quantum spins into fermions was done by employing a Jordan-Wigner transformation. Making use of translational invariance of the Hamiltonian, the operators were then respresented in the momentum space, where by using a Bogoliubov transformation, the Hamiltonian could be diagonalised. With the Hamiltonian written in the diagonal form, the dispersion relation could be determined and thus the energy spectrum found. After calculating the partition function and from there the free energy, it could be concluded that this model shows no phase transition occurring at finite temperature. However, the singularity at $T = 0$ implies that there indeed exists a phase transition, but only for a unique condition satisfied by the field strength and exchange energy, that allows the system to be in both a paramagnetic and a ferromagnetic state.

1 Introduction

The Ising model is a very successful mathematical model used mostly in statistical mechanics to represent the physics of atomic spins arranged in a lattice that are allowed to interact with their neighbors. The spins interact according to the Hamiltonian [1]:

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} S_i S_j - H \sum_i S_i,$$

where the first term is responsible for the interaction while the second one for the energy due to the applied field H. $\langle i, j \rangle$ denotes a sum over nearest neighbour spins and J is the exchange energy. It can be noted that due to the minus sign, the overall energy is lowered when neighbouring atomic spins are aligned, effect that is mostly due to the Pauli exclusion principle.

This model can be used for 1D chains or 2D lattices of spins and can also be extended to any other systems which display a simple interactive behaviour between nearest neighbours i.e. similar to the alignment of spins in a magnet. Two examples for this are the binary alloy which can be modelled by considering S_i to be 1 or -1 corresponding to the site i being occupied by an atom of type A or of type B, respectively, and the lattice gas model, for which $S_i = 1$ implies the site i is occupied, while $S_i = 0$ corresponds to a vacancy [1].

1.1 The One-Dimensional Transverse Ising Model

The quantum 1D Ising model describes a chain of quantum spins under the action of an applied field, that also interact with their nearest neighbours. A classical spin can be thought of as a vector pointing in a certain direction (up or down), which is still a fine representation even for quantum spins with extremely large spin number S. However, for a small S, the spins are in a superposition of the two states and therefore behave like entirely new objects with their own zero-point motion [2]. In 1D, it can be shown that spins with $S = 1/2$ behave like fermions.

In this essay, the Ising model in a transverse field will be analysed, as it is among the simplest classes of quantum models in statistical mechanics that can be solved exactly.

It consists of N one-half spins arranged in a cyclic chain with continuous boundary conditions, $S_{N+1} = S_1$ for all i with $1 \leq i \leq N$. The Hamiltonian is given by [3]:

$$\mathcal{H} = -J \sum_j S_j^x S_{j+1}^x - H \sum_j S_j^z, \tag{1}$$

where the choice of the axis is not entirely arbitrary, as this form will greatly simplify the forthcoming algebra. The Hamiltonian can be found in literature in different forms (regarding the choice of x, y and z), but they are all equivalent if one performs a canonical transformation [4].

1.2 Phase Transitions

Once knowing the form of the Hamiltonian of the system, one can determine the partition function, $\mathcal{Z} = \sum_{\{s\}} e^{-\beta\mathcal{H}}$,noting that $\{s\}$ traces over all possible states of the system. The Ising model is completely characterized by its free energy, $\mathcal{F} = -k_B \mathrm{T} \ln \mathcal{Z}$, and exactly calculating this energy is often regarded as the solution of the model.

Although most commonly used to describe transitions between solid, liquid or gaseous states of matter, a phase transition occurs whenever there is a singularity in the free energy or ones of its derivatives. This is characterised by a sharp transformation of the properties of the system, as a result of the change of external conditions. They can be divided into two different categories:

1. First-order phase transitions: they occur when there is a finite discontinuity in one or more of the first derivatives of the thermodynamic potential; these involve either an absorption or release of energy (like melting ice or boiling water);

2. Second-order phase transitions: they occur when the first derivatives are continuous but second derivatives are either discontinuous or infinite, which corresponds to a state of maximum order (like the transition from a paramagnet to a ferromagnet) [1].

2 Mathematical Framework

2.1 Creation and Annihilation Operators

In this section, only the creation and annihilation operators for fermions will be discussed, since the corresponding operators for bosons will not be used at all in the calculations that follow.

Suppose we have a fermionic system with single particle eigenstates $\phi_0(\mathbf{x})$, $\phi_1(\mathbf{x})$, ... , where $|n\rangle$ denotes there are n fermions in the specific level, with $n = 0, 1$. This state is assumed to be noramlized, $\langle n|n\rangle = 1$, and $|0\rangle$ represents the 'vacuum' state.

The destruction and creation operators acting on a certain level can be defined as [5]:

$$f\,|1\rangle = |0\rangle\,, \qquad f\,|0\rangle = 0,$$
$$f^\dagger\,|1\rangle = 0, \qquad f^\dagger\,|0\rangle = |1\rangle\,.$$

In the $|0\rangle = \binom{1}{0}$ and $|1\rangle = \binom{0}{1}$ basis, these operators can also be expressed in matrix form:

$$f = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

If we are now to consider two levels, ϕ_0 and ϕ_1, all there is to do is take the tensor product between the two single particle states, which for simplicity will be expressed with the help of

indices: for the bipartite state $|n_0, n_1\rangle$, f_0 and f_1 are destruction operators acting on the states $|n_0\rangle$ and $|n_1\rangle$, respectively, while $f_0^\dagger$ and $f_1^\dagger$ are construction operators.

Additionally, what now has to be considered in a multipartite system is the exclusion principle: interchanging the two fermions in a state must produce a sign change [5]. To better illustrate this, start by considering the state $|1,1\rangle$. Taking the fermion away from ϕ_1 implies:

$$|1,1\rangle \to |1,0\rangle = f_1 |1,1\rangle .$$

'Moving' the fermion from ϕ_0 to ϕ_1 is equivalent to annihilating the one from ϕ_0 and instead creating one in the state ϕ_1:

$$|1,0\rangle \to |0,1\rangle = f_1^\dagger f_0 |1,0\rangle .$$

The final step is to create a fermion in the state ϕ_0:

$$|0,1\rangle \to |1,1\rangle = f_0^\dagger |0,1\rangle .$$

Putting all these together, the antisymmetry requires that:

$$f_0^\dagger f_1^\dagger f_0 f_1 |1,1\rangle = -|1,1\rangle.$$

In order to account for this change of sign due to antisymmetry in a more general way, the operators can be defined by convention [6]:

$$f_i |n_0, n_1, ..., n_i, ...\rangle = (-1)^{\sum_{j<i} n_j} n_i |n_0, n_1, ..., 0_i, ...\rangle , \tag{2a}$$

$$f_i^\dagger |n_0, n_1, ..., n_i, ...\rangle = (-1)^{\sum_{j<i} n_j} (1 - n_i) |n_0, n_1, ..., 1_i, ...\rangle . \tag{2b}$$

Thus, any state of the system can be expressed as a series of creation operations on the vacuum:

$$|n_0, n_1, n_2, \ldots\rangle = \left(f_0^\dagger\right)^{n_0} \left(f_1^\dagger\right)^{n_1} \left(f_2^\dagger\right)^{n_2} \ldots |0\rangle .$$

The anticommutation relations can easily be checked:

$$\{f_i, f_j^\dagger\} = \delta_{ij}, \tag{3a}$$

$$\{f_i, f_j\} = \{f_i^\dagger, f_j^\dagger\} = 0. \tag{3b}$$

On a specific level, annihilating a fermion and then creating one in its place yields the number of particles. The number operator can thus be defined as:

$$F = \sum F_i = \sum_i f_i^\dagger f_i.$$

2.2 Jordan-Wigner Transformation

Unlike classical spins, quantum spins are extremely difficult objects to deal with as they do not behave as fermions or bosons. However, in one dimension, a mapping between spin $S = 1/2$ and fermions was found by Jordan and Wigner [2], which subsequently led to an analytic solution of the 1D Ising Model.

In order to map a state of N spins into the fermionic space, two conditions need to be satisfied: the same commutation and anticommutation relations have to be ensured and any state of the system, which is created from the vacuum state, must belong to both the spaces considered in the mapping [7]. The latter represents no concern, as it is automatically satisfied in our case.

They observed that the 'up' and 'down' states of a single spin can be thought of as empty and occupied fermion states in a gas: $|\downarrow\rangle \equiv |0\rangle$ and $|\uparrow\rangle \equiv f^\dagger |0\rangle$, which allowed the spin-rising and spin-lowering operators to be expressed as fermionic operators:

$$S^+ \equiv f^\dagger = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad \text{and} \qquad S^- \equiv f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Choosing the spins to be oriented in the z-direction, perpendicular to the $x-y$ plane in which they lie, and using the relations $S_\pm = S_x \pm iS_y$ and $[S_x, S_y] = iS_z$ from quantum mechanics, the one site spin operators can be written as:

$$S_x = \frac{1}{2}(S^+ + S^-) = \frac{1}{2}(f^\dagger + f), \tag{4a}$$

$$S_y = \frac{1}{2i}(S^+ - S^-) = \frac{1}{2i}(f^\dagger - f), \tag{4b}$$

$$S_z = \frac{1}{i}(S_x S_y - S_y S_x) = f^\dagger f - 1/2, \tag{4c}$$

where for S_z the fermionic anticommutation relations from Equation (3) were used. Performing the same algebra as in the previous section for all pairs x, y and z, the anticommutation of the spin operators is found to be:

$$\{S_a, S_b\} = \frac{1}{2}\delta_{ab}.$$

Using this result, it can be seen that for only one site the mapping from $(f, f^\dagger)$ to (S^-, S^+) obeys the anticommutation relation. For more sites, the way the operators have to be changed is by applying the single site operator on the wanted site and then taking the tensor product with the identity matrix $\mathbb{1}_2$ on all the other sites, i.e. $S_a(\text{on site } j) = \mathbb{1} \otimes \mathbb{1} \otimes \cdots \otimes S_a(\text{site } j) \otimes \cdots \otimes \mathbb{1}$, where S_a can be any spin operator. However, when mapping more spin sites, it can be seen that independent spin operators commute while independent fermionic operators anticommute.

Jordan and Wigner discovered a way to fix this in one dimension, namely by attaching to the fermions a phase factor called a string: $S_j^+ = f_j^\dagger \, e^{i\phi_j}$, where the phase contains the sum over all

fermionic occupancies at sites to the left of j: $\phi_j = \pi \sum_{l<j} n_l$.

An intuitive way of justifying why a phase factor like this would be added is by thinking that it cancels out the effect of the antisymmetry of the fermions, which also have a similar phase factor attached to the operators by convention (as it can be seen in Equations (2)).

The complete Jordan-Wignar transformation, with S_j^+ and S_j^- chosen in such a way that they are Hermitian conjugates, is then [2]:

$$S_j^z = f_j^\dagger f_j - 1/2, \tag{5a}$$
$$S_j^+ = f_j^\dagger \, e^{i\pi \sum_{l<j} n_l}, \tag{5b}$$
$$S_j^- = f_j \, e^{-i\pi \sum_{l<j} n_l}. \tag{5c}$$

An importnat thing to notice now is that the operator $e^{i\pi n_j}$ anticommutes with the fermionic one, since when the state j is vacant the calculation is trivial and when it is occupied, the operator f_j reduces n_j from 1 to 0, while $e^{i\pi n_j}$ adds a phase of $e^{i\pi} = -1$ when acting on the right. For the conjugate expression applying the same reasoning gives:

$$\{e^{i\pi n_j}, f_j\} = \{e^{i\pi n_j}, f_j^\dagger\} = 0. \tag{6}$$

Other relations between the string and the fermionic operators that are relevant and which can be checked by direct calculation are:

$$\{e^{i\phi_j}, f_l^{(\dagger)}\} = 0 \qquad (l < j), \tag{7a}$$
$$[\, e^{i\phi_j}, f_l^{(\dagger)}\,] = 0 \qquad (l \geq j). \tag{7b}$$

2.3 Bogoliubov Transformation

In the fermionic mode, usually in the momentum space, a widely used method of diagonalising Hamiltonians is by employing a Bogoliugov transformation. The way it works is by making a linear combination of the operators f_k and $f_k^\dagger$ such that the new operators η_k and $\eta_k^\dagger$ remove from the Hamiltonian the terms that do not conserve the particle number [4].

Mathematically, the transformation is as follows:

$$\begin{pmatrix} \eta_k \\ \eta_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & i\,v_k \\ i\,v_k & u_k \end{pmatrix} \begin{pmatrix} f_k \\ f_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k f_k + i\,v_k f_{-k}^\dagger \\ i\,v_k f_k + u_k f_{-k}^\dagger \end{pmatrix}, \tag{8}$$

provided that it satisfies three criteria:

1. The new set of operators η_k and $\eta_k^\dagger$ are fermionic (i.e. satisfy the same commutation and anticommutation relations as the fermions: Equation (3));
2. They diagonalise the Hamiltonian: $\mathcal{H} = U\,D\,U^\dagger$, where D is the diagonal matrix consisting of the eigenvalues of the matrix $\mathcal{H}$;
3. u_k and v_k are chosen to be real such that the transformation U is unitary: $u_k^2 + v_k^2 = 1$ [8].

3 Exact Solution of the 1D Quantum Ising Model

Solving the Transverse Ising Model comes down to determining the dispersion relation of the Hamiltonian, i.e. writing the Hamiltonian in a form 'energy(α) $\times$ number of particles α', while summing over all possible 'α'-s. We therefore start with the Hamiltonian given in Equation (1).

Expressing the first sum $S_j^x\, S_{j+1}^x$ in terms of the fermionic operators requires the use of the Jordan-Wigner transformation explained in section 2.2. Each of the two terms S^x can be written in terms of the operators $S^{\pm}$, using the first part of equation (4a) for a single state and then expressing $S^{\pm}$ from equations (5b) and (5c).

This will give rise to a sum of 4 terms operating on either the site j or $j+1$. From here, in order to obtain a sum that does not depend on the string, but only on the fermionic operators, the string operators acting on the $j+1$ have to be rewritten as:

$$e^{i\pi \sum_{l<j+1} n_l} = e^{i\pi \sum_{l<j} n_l}\, e^{i\pi n_j}.$$

At this point, all 4 terms in the sum should be of the form $f_j^{(\dagger)}\, e^{\pm i\pi \sum n_l}\, f_{j+1}^{(\dagger)}\, e^{\pm i\pi \sum n_l}\, e^{\pm i\pi n_j}$, where $^{(\dagger)}$ and $\pm$ do not imply corresponding terms, but rather arbitrary combinations of them.

Using the commutation relation given in Equation (7b), each of the second and the third terms can be interchanged so that the string operators are applied consecutively. In this case, they can either cancel each other out for the terms $e^{\pm i\pi \sum n_l}\, e^{\mp i\pi \sum n_l}$ or simply become 1 for the terms $e^{\pm 2i\pi \sum n_l}$. In either case, the remaining terms are now of the form $f_j^{(\dagger)}\, f_{j+1}^{(\dagger)}\, e^{\pm i\pi n_j}$.

From Equation (6) the action upon the $e^{\pm i\pi n_j}$ operator can be summed as: $f_j\, e^{\pm i\pi n_j} = -f_j$ and $e^{\pm i\pi n_j}\, f_j^{\dagger} = -f_j^{\dagger}$. Keeping these in mind, the next goal is to reposition the terms so that the fermionic operators are right next to the phase, which can be done by swapping $f_j^{(\dagger)}$ and $f_{j+1}^{(\dagger)}$ with the use of the anticommutation relations given in Equation (3). This now gives:

$$S_j^x\, S_{j+1}^x = \frac{1}{4}(f_j^{\dagger}\, f_{j+1}^{\dagger} + f_j^{\dagger}\, f_{j+1} + f_{j+1}^{\dagger}\, f_j + f_{j+1}\, f_j). \tag{9}$$

The second part of the Hamiltonian is simply $-H \sum_j (f_j^{\dagger}\, f_j - 1/2)$.

Since the Hamiltonian has translational invariance, it can be diagonalised in the momentum space [9]. Considering the spins as being aligned in a ring with continuous boundaries and lattice space a, the Fourier transform of the fermionic operators becomes:

$$f_k = \frac{1}{\sqrt{N}} \sum_{n=1}^{N} f_n\, e^{inka},$$

with $k \in (-\frac{\pi}{a}, \frac{\pi}{a})$. The anticommutation relations are preserved in the k-space:

$$\{f_k, f_{k'}^{\dagger}\} = \delta_{k,k'} \quad \text{and} \quad \{f_k, f_{k'}\} = \{f_k^{\dagger}, f_{k'}^{\dagger}\} = 0. \tag{10}$$

The corresponding transformations for fermionic operators in real space are:

$$f_j^{(\dagger)} = \frac{1}{\sqrt{N}} \sum_k f_k^{(\dagger)} e^{\mp ikja} \qquad \text{and} \qquad f_{j+1}^{(\dagger)} = \frac{1}{\sqrt{N}} \sum_{k'} f_{k'}^{(\dagger)} e^{\mp ik'(j+1)a}, \tag{11}$$

where the $-$ sign corresponds to the annihilation operators and the $+$ sign to the creation ones.

Substituting the operators from (11) into Equation (9) and then putting back everything into the Hamiltonian leads to a long summation over j, k and k'. This can be simplified using the fact that $\sum_{n=1}^{N} e^{inka} = N\,\delta_{k,0}$ (from the definition of the Dirac delta), since by summing over j and then k', the latter will be replaced with $\pm k$. The Hamiltonian becomes:

$$\mathcal{H} = -\frac{J}{4} \sum_k f_k^\dagger\, f_{-k}^\dagger\, e^{-ika} + f_k^\dagger\, f_k\, e^{-ika} + f_k^\dagger\, f_k\, e^{ika} + f_{-k}\, f_k\, e^{ika} - H \sum_k f_k^\dagger\, f_k + \frac{H\,N}{2}.$$

The aim at this point is to rewrite $\mathcal{H}$ in such a way that it can be expressed as a sum of terms combining $f_{\pm k}^{(\dagger)}\, f_{\pm k}^{(\dagger)}$. This can be done by noticing the fact that on an even interval, summing over k is equivalent to summing over $-k$. Thus writing $f_k^\dagger\, f_{-k}^\dagger e^{-ika}$ as $\frac{1}{2}(f_k^\dagger\, f_{-k}^\dagger + f_k^\dagger\, f_{-k}^\dagger)e^{-ika}$ and then summing the first part from $-\pi/a$ to π/a and the second one from π/a to $-\pi/a$ will give rise to a term e^{+ika} which after taking $f_k^\dagger\, f_{-k}^\dagger$ as common factor, gives $(e^{-ika} - e^{ika})/2 = -i\,\sin(ka)$.

Applying the same reasoning for $f_{-k}\, f_k$ and $f_k^\dagger\, f_k$ and using the Equation (10) to swap the terms around, the Hamiltonian, $\mathcal{H} = \sum_k \mathcal{H}_k$, can be expressed in the form:

$$\mathcal{H}_k = -\begin{pmatrix} f_k^\dagger & f_{-k} \end{pmatrix} \begin{pmatrix} \bar{\lambda}\cos ka + \frac{H}{2} & -i\,\bar{\lambda}\sin ka \\ i\,\bar{\lambda}\sin ka & -\bar{\lambda}\cos ka - \frac{H}{2} \end{pmatrix} \begin{pmatrix} f_k \\ f_{-k}^\dagger \end{pmatrix},$$

where $\bar{\lambda} = -J/4$ and the term $\frac{HN}{2}$ is canceled out by the sum $\sum_k 1 \cdot \frac{H}{2}$, which is left after using the anticommutation from the first part of Equation (10): $f_{-k}^\dagger\, f_{-k} = 1 - f_{-k}\, f_{-k}^\dagger$.

Performing the Bogoliubov transformation described in Section 2.3 on the Hamiltonian gives:

$$\mathcal{H}_k = -\begin{pmatrix} \eta_k^\dagger & \eta_{-k} \end{pmatrix} \begin{pmatrix} u_k & i\,v_k \\ i\,v_k & u_k \end{pmatrix} \begin{pmatrix} \bar{\lambda}\cos ka + \frac{H}{2} & -i\,\bar{\lambda}\sin ka \\ i\,\bar{\lambda}\sin ka & -\bar{\lambda}\cos ka - \frac{H}{2} \end{pmatrix} \begin{pmatrix} u_k & -i\,v_k \\ -i\,v_k & u_k \end{pmatrix} \begin{pmatrix} \eta_k \\ \eta_{-k}^\dagger \end{pmatrix}.$$

Since this set of operators are chosen such that they diagonalise the Hamiltonian, it is required that $(u_k, -i\,v_k)^T$ is an eigenvector of $\mathcal{H}$ [4]:

$$\begin{pmatrix} \bar{\lambda}\cos ka + \frac{H}{2} & -i\,\bar{\lambda}\sin ka \\ i\,\bar{\lambda}\sin ka & -\bar{\lambda}\cos ka - \frac{H}{2} \end{pmatrix} \begin{pmatrix} u_k \\ -i\,v_k \end{pmatrix} = -\varepsilon_k \begin{pmatrix} u_k \\ -i\,v_k \end{pmatrix}.$$

Any 2×2 Hermitian matrix can be written as a linear combination of Pauli matrices with real coefficients: $A = a_0\mathbb{1} + a_x\sigma_x + a_y\sigma_y + a_z\sigma_z$, as together with $\mathbb{1}_2$ they constitute a complete basis:

$$\mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; \quad \hat{\sigma}^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad \hat{\sigma}^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \quad \hat{\sigma}^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

In this basis, an important property is that the eigenvalues of the matrix can be written as $\varepsilon_\pm = a_0 \pm \sqrt{a_x^2 + a_y^2 + a_z^2}$. For $a_0 = a_x = 0$, $a_y = \bar{\lambda} \sin ka$ and $a_z = \bar{\lambda} \cos ka + \frac{H}{2}$, this gives:

$$\varepsilon_k = \pm\sqrt{\bar{\lambda}^2 + \bar{\lambda} H \cos ka + H^2/4}. \tag{12}$$

The diagonalised Hamiltonian is now:

$$\mathcal{H} = -\sum_k \begin{pmatrix} \eta_k^\dagger & \eta_{-k} \end{pmatrix} \begin{pmatrix} -\varepsilon_k & 0 \\ 0 & \varepsilon_k \end{pmatrix} \begin{pmatrix} \eta_k \\ \eta_{-k}^\dagger \end{pmatrix} = 2\sum_k \varepsilon_k \eta_k^\dagger \eta_k - \sum_k \varepsilon_k,$$

where for the last part the anticommutation relation $\{\eta_k, \eta_k^\dagger\} = 1$ was used, along with the fact that $\varepsilon_{-k} = \varepsilon_k$ from Equation (12), since the summations over k and $-k$ are equivalent. The ground-state energy of the system is $E_0 = -\sum_k \varepsilon_k$.

4 Phase Transitions in the 1D Quantum Ising Model

The first step in finding whether there is a phase transition or not is to calculate the free energy as described previously in Section 1.2. Since the spin system is equivalent to a fermionic one and the partition function sums over all the states, the 'number of particle' $\eta_k^\dagger \eta_k$ can only take vales 0 or 1. The partition function can be written as:

$$\mathcal{Z} = \sum_{\{s\}} e^{-\beta\mathcal{H}} = e^{-\beta E_0} \sum_{\{s\}} \prod_k e^{-2\beta\varepsilon_k \eta_k^\dagger \eta_k} = e^{-\beta E_0} \prod_k (e^{-2\beta\varepsilon_k} + 1).$$

The free energy is therefore:

$$\mathcal{F} = -k_B T \ln \mathcal{Z} = -k_B T \Big[-\beta E_0 + \sum_k \ln\left(e^{-2\beta\varepsilon_k} + 1\right) \Big],$$

where the sum can be approximated with an integral of the form $\int_{-\pi}^{\pi} dk \, \frac{N}{2\pi} \ln(...)$ [3].

By writing the sum inside the logarithm in terms of $\cosh(\beta\varepsilon_k)$ and noting that from the definition of the ground-state energy $E_0 = -\int_\pi^\pi dk \, \frac{N}{2\pi}$, the final form of the free energy is found to be:

$$\mathcal{F} = -N k_B T \Big[\ln 2 + \frac{1}{\pi} \int_0^\pi dk \, \ln\big(\cosh(\beta\varepsilon_k)\big) \Big].$$

Solving the integral numerically and plotting the function $\mathcal{F}(T)$ for different values of $\bar{\lambda}$ and H, it can be seen that the free energy does not show any singularities at any finite temperature, although it diverges at $T = 0$. The free energy as a function of T, together with the energy spectrum given in Equation (12), are illustrated in Figure 1.

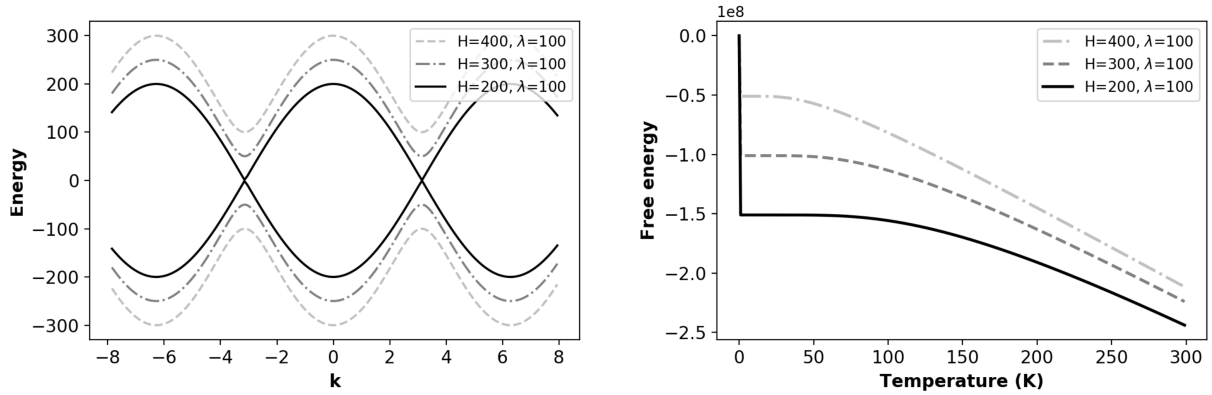


Figure 1: For different values of $\bar{\lambda}$ and H, the energy spectrum of the system (on the left) and the variation of the free energy with temperature (on the right) were plotted using Python.

5 Discussion

From the energy spectrum plotted above, it can be seen that there are two energy bands whose behaviour depend solely on the parameters $\bar{\lambda}$ and H that characterise the Ising model, namely the strength or the neighbouring interaction and of the applied field, respectively. By direct evaluation or from the plot, it can be observed that the two bands touch when the condition $H = 2\bar{\lambda} = J/2$ is satisfied. Physically, the points where the bands intersect, called 'critical points', correspond to the level crossing that marks the boundary between the ferromagnetic and paramagnetic phases [9].

For $J >> H$, the nearest-neighbour coupling term dominates and therefore the ground state is expected to be influenced mostly by the exclusion principle, which corresponds to consecutive spins being antiparallel with each other. This is also the description of a paramagnet. For $J << H$, the external field dominates and therefore the spins are all expected to be aligned with the field, description that full characterises a ferromagnet. However, going closer to the critical points, the physics can only be described using a continuum quantum field theory [9].

One of the main areas of research in condensed matter physics is focused on quantum phase transitions, while the 1D transverse Ising model is one of the simplest solvable representations of an interacting system that allows the study of these transitions.

As seen in the right plot from Figure 1, the free energy does not present any discontinuities at $T \neq 0$, which means that physically, there are no phase transitions occuring at finite temperature. However, the singularity at $T = 0$ together with the condition between $\bar{\lambda}$ and H at the critical point show the theoretical existence of a phase transition between a paramagnet and a ferromagnet.

The transverse model was originally introduced as a pseudospin model for hydrogen-bonded ferroelectrics in the order-disorder phenomenon with tunneling effects and it is also used nowadays to analyse real anisotropic magnetic materials in a transverse field [10]. Another major application of the one-dimensional quantum Ising model is the possibility of mapping the two-dimensional classical Ising model into it. More generally, it was shown that there exists a correspondence between any $(N+1)$- dimensional classical Ising system and a N-dimensional quantum one [9].

6 Conclusion

The dispersion relation, energy spectrum and free energy of the one-dimensional Ising model in a transverse field were determined and analysed. This was achieved by mapping the spin operators into fermionic ones and then using different mathematical techniques so that the Hamiltonian could be written in a diagonal form. After evaluating the energy bands, a critical point of paramagnetic-ferromagnetic coexistance was found, whose detailed description can only be achieved using quantum field theory [9]. However, as the free energy was shown not to display any discontinuities in itself or its derivatives, it could be justified why there are no phase transitions occurring at finite temperature in this model.

References

[1] J. Yeomans, *Statistical Mechanics of Phase Transitions*. Clarendon Press, 1992.

[2] P. Coleman, *Simple examples of second quantization*, pp. 71–94. Cambridge University Press, 2015.

[3] P. Pfeuty, "The one-dimensional ising model with a transverse field," *Annals of Physics*, vol. 57, pp. 79–90, Mar. 1970.

[4] S. Suzuki, J. Inoue, and B. Chakrabarti, *Transverse Ising Chain (Pure System)*, pp. 13–46. jan 2013.

[5] F. Porter, "Second quantization." Unpublished.

[6] H. Johannesson, "second quantization (the occupation-number representation)." Unpublished, feb 2013.

[7] M. Steudtner and S. Wehner, "Fermion-to-qubit mappings with varying resource requirements for quantum simulation," *New Journal of Physics*, vol. 20, pp. 1–2, jun 2018.

[8] N. Bogoluibov and V. Tomachev, *A New Method in the Theory of Superconductivity*. Literary Licensing, LLC, 2013.

[9] A. Tan, "Quantum ising models." Unpublished, dec 2018.

[10] M. A. Neto and J. R. de Sousa, "Phase diagrams of the transverse ising antiferromagnet in the presence of the longitudinal magnetic field," *Physica A: Statistical Mechanics and its Applications*, vol. 392, no. 1, pp. 1 – 6, 2013.